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On Extrinsic Information of Good Codes Operating over Discrete Memoryless Channels

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Abstract: We show that the Extrinsic Information about the coded bits of any good (capacity achieving) code operating over a wide class of discrete memoryless channels (DMC) is zero when channel capacity is below the code rate and positive constant otherwise, that is, the Extrinsic Information Transfer (EXIT) chart is a step function of channel quality, for any capacity achieving code. It follows that, for a common class of iterative receivers where the error correcting decoder must operate at first iteration at rate above capacity (such as in turbo equalization, turbo channel estimation, parallel and serial concatenated coding and the like), classical good codes which achieve capacity over the DMC are not effective and should be replaced by different new ones. Another meaning of the results is that a good code operating at rate above channel capacity falls apart into its individual transmitted symbols in the sense that all the information about a coded transmitted symbol is contained in the corresponding received symbol and no information about it can be inferred from the other received symbols.

This report comprises two stand-alone parts. Part 1 treats the binary input additive white Gaussian noise channel. Part 2 extends the results to the symmetric binary channel and to the binary erasure channel and provides an heuristic extension to wider class of channel models such as common fading models combined with QPSK and other inputs which might be rigorized by further work.